```
import pandas as pd
import matplotlib.pyplot as plt
from sqlalchemy import create_engine


print('-------------------------------------------
-----------------------------')
print('\t\t\t COVID ANALYSIS \t\t\t')
print('-------------------------------------------
-----------------------------')


engine=create_engine('mysql+pymysql://r
oot:abcde@localhost:3306/MYDB')
df1=pd.read_sql('SELECT*FROM
ACTIVE_CASES',engine)
d1=pd.DataFrame(df1)
df2=pd.read_sql('SELECT*FROM
CONFIRMED_CASES',engine)
d2=pd.DataFrame(df2)
```

```
df3=pd.read_sql('SELECT*FROM
DEATHS',engine)
d3=pd.DataFrame(df3)
df4=pd.read_sql('SELECT*FROM
RECOVERIES',engine)
d4=pd.DataFrame(df4)


z1=d1['END_I'].sum()
z2=d2['END_I'].sum()
z3=d3['END_I'].sum()
z4=d4['END_I'].sum()
z5=d1['END_II'].sum()
z6=d2['END_II'].sum()
z7=d3['END_II'].sum()
z8=d4['END_II'].sum()
x=['ACTIVE','CONFIRMED','DEATHS','
RECOVERIES']
```

```
a=0
while a!=-1:
    print('\nMODULES AVAILABLE TO VIEW:\n')
    print('1. FIRST AND SECOND WAVE COMPARISON')
    print('2. ACTIVE CASES')
    print('3. CONFIRMED CASES')
    print('4. DEATHS')
    print('5. RECOVERIES')
    print('6. GENDER-WISE IMPACT')
    print()
    a=int(input('Choose module to view:'))
    if a==2:
        print('\n\t\t\t ACTIVE CASES\n',df1)
        print('\t\t\t _____\t\t\t\n')
        b=input('Do you want to view the graph?(YES/NO):')
```

```
        if b=='yes':

d1.plot(kind='barh',x='CITY',color=['b','r','
m','gold'])
            plt.legend()
            plt.title('ACTIVE CASES')
            plt.ylabel('CITIES')
            plt.xlabel('NO. OF CASES')
            plt.show()
            print()
        else:
            exit
    elif a==3:
        print('\t\t\t CONFIRMED
CASES\n',df2)
        print('\t\t\t _____\t\t\t\n')
        b=input('Do you want to view the
graph?(YES/NO):')
```

```
        if b=='yes':

d2.plot(kind='barh',x='CITY',color=['b','r','
m','gold'])
            plt.legend()
            plt.title('CONFIRMED CASES')
            plt.ylabel('CITIES')
            plt.xlabel('NO. OF CASES')
            plt.show()
            print()
        else:
            exit
    elif a==4:
        print('\t\t\t DEATHS\n',df3)
        print('\t\t\t _____\t\t\t\n')
        b=input('Do you want to view the
graph?(YES/NO):')
        if b=='yes':
```

```
d3.plot(kind='barh',x='CITY',color=['b','r','
m','gold'])
            plt.legend()
            plt.title('DEATHS')
            plt.ylabel('CITIES')
            plt.xlabel('NO. OF CASES')
            plt.show()
            print()
        else:
            exit
    elif a==5:
        print('\t\t\t RECOVERIES\n',df4)
        print('\t\t\t _____\t\t\t\n')
        b=input('Do you want to view the
graph?(YES/NO):')
        if b=='yes':
```

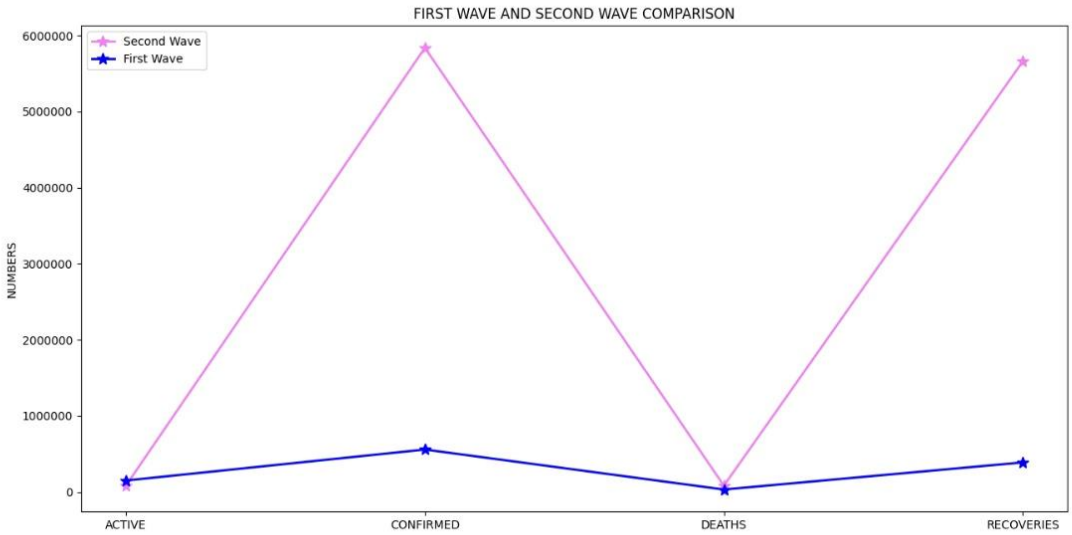
```
d4.plot(kind='barh',x='CITY',color=['b','r','
m','gold'])
            plt.legend()
            plt.title('RECOVERIES')
            plt.ylabel('CITIES')
            plt.xlabel('NO. OF CASES')
            plt.show()
            print()
        else:
             exit
    elif a==6:
        gender=['Male','Female']
        n=[73797,39063]
        color=['Cyan','Red']

plt.pie(n,labels=gender,colors=color,autop
ct='%.2f',explode=[0.1,0.1],shadow=True)
```

```
        plt.title('Gender-wise Impact')
        plt.legend()
        plt.show()
    elif a==1:
        plt.ticklabel_format(style='plain')

plt.plot(x,[z5,z6,z7,z8],marker='*',markers
ize=10,linewidth=2,color='violet')

plt.plot(x,[z1,z2,z3,z4],marker='*',markers
ize=10,linewidth=2,color='blue')
        plt.legend(['Second Wave','First
Wave'])
        plt.ylabel('NUMBERS')
        plt.title('FIRST WAVE AND
SECOND WAVE COMPARISON')
        plt.show()
    else:
```

```
        break
print('\nThank you for viewing out project!')
print('Let us fight against and sustain ourselves amidst these tough times by wearing a mask and following covid protocols, and be kinder to everyone around us.')
```

OUTPUTS

```
--------------------------------------------------------------------------
                          COVID ANALYSIS
--------------------------------------------------------------------------

MODULES AVAILABLE TO VIEW:

1. FIRST AND SECOND WAVE COMPARISON
2. ACTIVE CASES
3. CONFIRMED CASES
4. DEATHS
5. RECOVERIES
6. GENDER-WISE IMPACT

Choose module to view:1
```

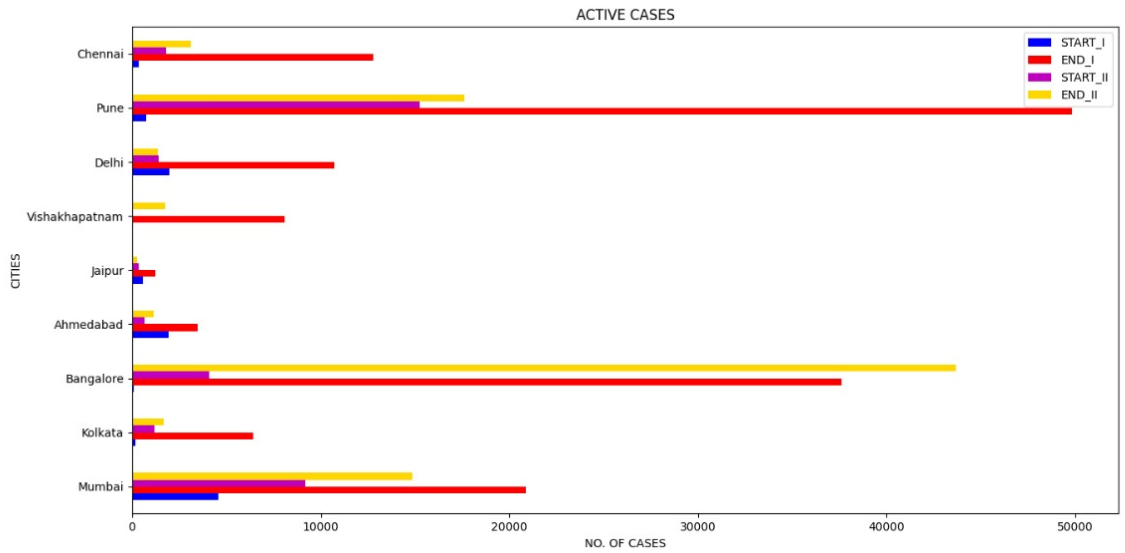


```
MODULES AVAILABLE TO VIEW:

1. FIRST AND SECOND WAVE COMPARISON
2. ACTIVE CASES
3. CONFIRMED CASES
4. DEATHS
5. RECOVERIES
6. GENDER-WISE IMPACT

Choose module to view:2
```

ACTIVE CASES

	CITY	START_I	END_I	START_II	END_II
0	Mumbai	4562	20857	9164	14873
1	Kolkata	184	6422	1188	1656
2	Bangalore	71	37622	4093	43699
3	Ahmedabad	1937	3452	642	1143
4	Jaipur	565	1227	335	249
5	Vishakhapatnam	3	8074	24	1748
6	Delhi	1987	10705	1404	1379
7	Pune	766	49821	15248	17610
8	Chennai	355	12768	1786	3121

```
Do you want to view the graph?(YES/NO):yes
```

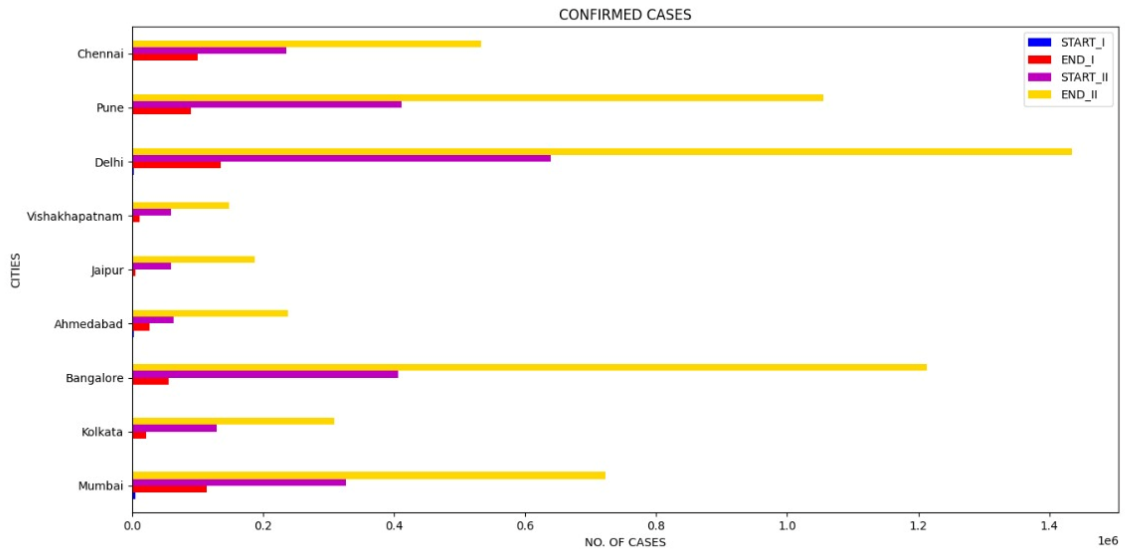


```
MODULES AVAILABLE TO VIEW:

1. FIRST AND SECOND WAVE COMPARISON
2. ACTIVE CASES
3. CONFIRMED CASES
4. DEATHS
5. RECOVERIES
6. GENDER-WISE IMPACT

Choose module to view:3
```

CONFIRMED CASES

	CITY	START_I	END_I	START_II	END_II
0	Mumbai	5407	114284	326772	722222
1	Kolkata	184	20969	129481	308091
2	Bangalore	129	55544	405847	1213559
3	Ahmedabad	2181	26517	62882	237681
4	Jaipur	808	5418	59562	187196
5	Vishakhapatnam	22	10765	60029	148622
6	Delhi	2918	135598	639464	1434188
7	Pune	1052	89231	410699	1055057
8	Chennai	528	99794	235721	532529

```
Do you want to view the graph?(YES/NO):yes
```

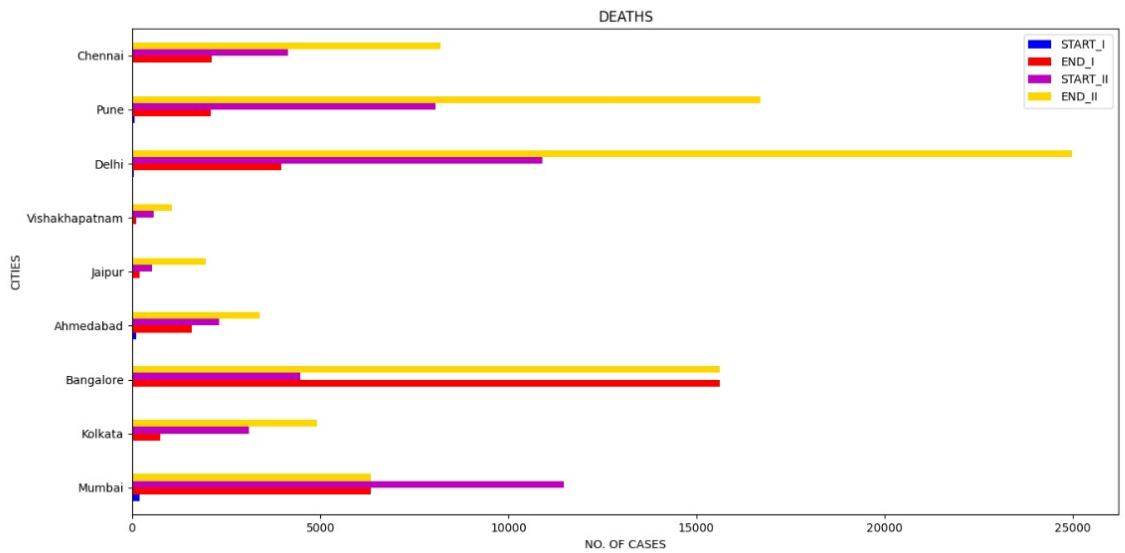


```
MODULES AVAILABLE TO VIEW:

1. FIRST AND SECOND WAVE COMPARISON
2. ACTIVE CASES
3. CONFIRMED CASES
4. DEATHS
5. RECOVERIES
6. GENDER-WISE IMPACT

Choose module to view:4
```

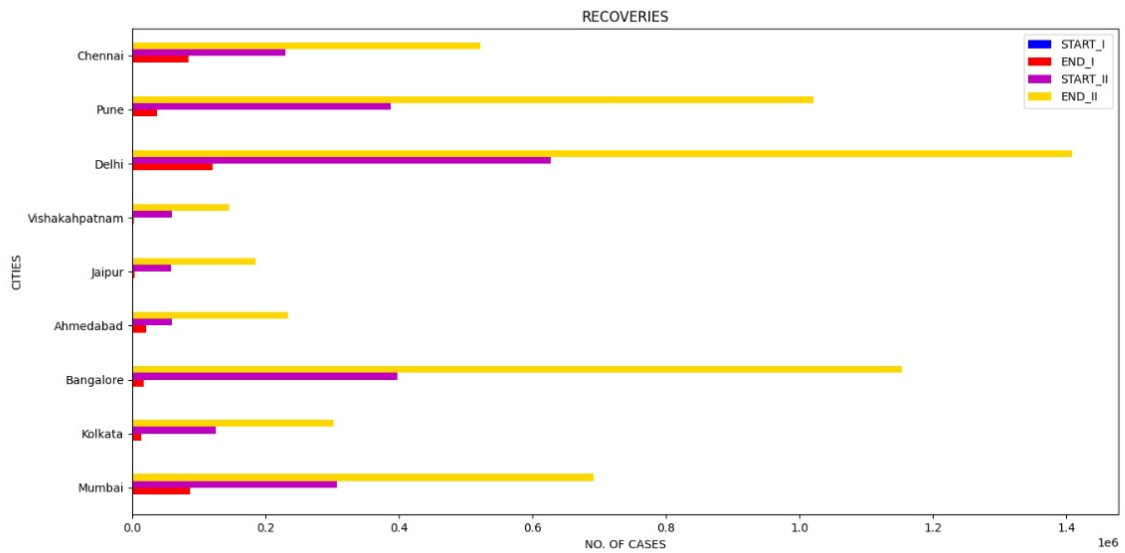
DEATHS

	CITY	START_I	END_I	START_II	END_II
0	Mumbai	204	6353	11479	6353
1	Kolkata	0	739	3101	4915
2	Bangalore	5	15626	4483	15626
3	Ahmedabad	104	1597	2314	3403
4	Jaipur	21	188	519	1970
5	Vishakhapatnam	0	105	567	1052
6	Delhi	54	3963	10911	24977
7	Pune	76	2099	8063	16693
8	Chennai	12	2110	4152	8187

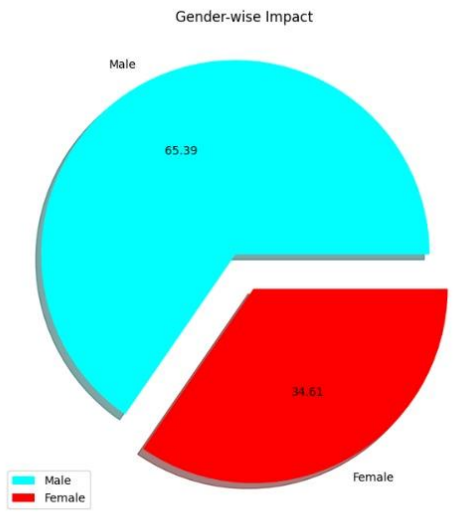
```
Do you want to view the graph?(YES/NO):yes
```



```
MODULES AVAILABLE TO VIEW:

1. FIRST AND SECOND WAVE COMPARISON
2. ACTIVE CASES
3. CONFIRMED CASES
4. DEATHS
5. RECOVERIES
6. GENDER-WISE IMPACT

Choose module to view:5
                          RECOVERIES
              CITY  START_I   END_I  START_II   END_II
0           Mumbai      641   87074    306129   691898
1          Kolkata        0   13808    125192   301520
2        Bangalore       53   16896    397271  1154234
3        Ahmedabad      140   21468     59926   233135
4           Jaipur      222    4003     58708   184977
5  Vishakahpatnam       19    2586     59438   145822
6            Delhi      877  120930    627149  1407832
7             Pune      210   37311    387388  1020754
8          Chennai      161   84916    229783   521221

Do you want to view the graph?(YES/NO):yes
```



```
MODULES AVAILABLE TO VIEW:

1. FIRST AND SECOND WAVE COMPARISON
2. ACTIVE CASES
3. CONFIRMED CASES
4. DEATHS
5. RECOVERIES
6. GENDER-WISE IMPACT

Choose module to view:6
```




```
MODULES AVAILABLE TO VIEW:

1. FIRST AND SECOND WAVE COMPARISON
2. ACTIVE CASES
3. CONFIRMED CASES
4. DEATHS
5. RECOVERIES
6. GENDER-WISE IMPACT

Choose module to view:7

Thank you for viewing out project!
Let us fight against and sustain ourselves amidst these tough times by wearing a mask and
following covid protocols, and be kinder to everyone around us.
>>>
```